RM09

Numeracy I: Ratio, Proportion, Partnership, Percentage, Profit & Loss, Interest

3-4 questions a paper, the largest single block in the reasoning section - the bank carries 70 questions for this chapter. Interest is the biggest sub-topic: the CI-minus-SI difference was asked in RAS Pre 2018, 2021, 2023 AND 2024. Milk-and-water mixtures came in 2016 and again in 2024.

This is the arithmetic block of the reasoning paper - the numbers you did in school, asked in one line with four options. Nothing here is hard; everything here is fast if you carry the shortcut formula and slow if you set up equations. The difference between compound and simple interest alone has come in four consecutive papers.

Numeracy I

CI minus SI (the star)

- 2 years: P(R/100)^2
- 3 years: P x R^2 x (300+R)/10^6
- Reverse: P = D x (100/R)^2
- Reverse: R = 100 x sqrt(D/P)

Interest

- SI = PRT/100
- A = P(1 + R/100)^n
- Half-yearly: rate R/2, periods 2n
- CI doubling in t years = 4x in 2t

Ratio & proportion

- Share = (own part / total parts) x amount
- Combine a:b and b:c by matching b
- Mean proportional = sqrt(ab)
- Third proportional = b^2/a

Mixture & alligation

- cheaper : dearer = (d - m) : (m - c)
- Water is taken at zero cost
- Replacement: x(1 - y/x)^n
- Equal quantities: add the fractions

Percentage

- Successive: a + b + ab/100
- x% up then x% down = x^2/100 fall
- Price up x% -> cut use by 100x/(100+x)%
- Growth P(1+r/100)^n, fall P(1-r/100)^n

Profit, loss & discount

- Percentages are always on the COST price
- SP = MP(100 - d)/100
- Two discounts: d1 + d2 - d1d2/100
- Same SP, +x% and -x% = x^2/100 loss

Partnership

- Profit ratio = capital x time
- Working partner's salary comes out first
- Withdrawal changes the time, not the ratio of capital
- Joining late = fewer months

How this block is asked, and the two habits that save the marks

Every question in this chapter is a one-liner with four options, and every one of them has a shortcut. The examiner is not testing algebra; he is testing whether you carry the formula. So the whole preparation is: know which family the question belongs to, write the one formula for that family, put the numbers in. Setting up 'let the number be x' costs you thirty seconds you do not have.

- Ratio, proportion and the division of an amount into shares.
- Mixture and alligation - the milk-and-water family.
- Partnership - profit split by capital multiplied by time.
- Percentage, successive change, population growth and depreciation.
- Profit, loss, marked price and discount.
- Simple interest, compound interest and instalments.
- The difference between CI and SI - a family of its own, and the most repeated one.

Percentage to fraction - learn these and half the arithmetic disappears

Percentage	Fraction	Percentage / Fraction
50%	1/2	20% = 1/5
33 1/3%	1/3	16 2/3% = 1/6
25%	1/4	12 1/2% = 1/8
10%	1/10	6 1/4% = 1/16
8 1/3%	1/12	5% = 1/20
66 2/3%	2/3	37 1/2% = 3/8

YAAD RAKHO

Second habit: always ask 'percentage of WHAT'. Profit and loss percentages are on the cost price. Discount is on the marked price. A percentage rise and the matching percentage fall are on different bases - that is why they never cancel. Ninety per cent of the wrong answers in this chapter are right arithmetic on the wrong base.

Percentage and successive change

Percentage questions come in three shapes: plain conversion, successive change, and the 'A is x% more than B' comparison. The successive-change shape is the one that traps people, because two changes of a% and b% never simply add. The second change acts on the already-changed amount, so the net effect is a + b + ab/100, and you write a fall with a minus sign.

The percentage rules worth memorising

Situation	Formula	Note
a% change then b% change	a + b + ab/100	Use a minus sign for a fall
x% rise then x% fall	Net fall of x^2/100 per cent	Never zero
Price rises x%, expenditure fixed	Cut consumption by 100x/(100 + x) per cent	x = 25 gives 20%
A is x% more than B	B is 100x/(100 + x) per cent less than A	Different base, different number
A is x% less than B	B is 100x/(100 - x) per cent more than A	-
Growth for n years at r%	P(1 + r/100)^n	Depreciation: P(1 - r/100)^n

WORKED EXAMPLE 1. (a) A price is increased by 20% and the new price is then increased by a further 10% - what is the net increase? (b) A salary of Rs 25,000 is raised by 10% and the raised salary is then cut by 10% - what is the final salary? (c) If the price of a commodity rises by 25%, by what percentage must a family cut its consumption to keep the expenditure unchanged?

- (a) Net = a + b + ab/100 = 20 + 10 + (20 x 10)/100 = 30 + 2 = 32%.
- (a) Check the long way: 100 becomes 120, and 120 becomes 132. Yes, 32%.
- (b) Net = 10 - 10 - (10 x 10)/100 = -1%, a fall of 1%, i.e. Rs 250.
- (b) Final salary = 25000 - 250 = Rs 24,750. Long way: 25000 x 1.10 = 27,500, then 27500 x 0.90 = 24,750.
- (c) Required cut = 100x/(100 + x) = 100 x 25/125 = 20%.

- (c) Check: 100 units at Re 1 cost Rs 100; at Rs 1.25 the same Rs 100 buys only 80 units - a 20% cut.

TRAP

A 20% rise followed by a 20% fall is NOT 'no change'. It is a 4% fall, because the fall is computed on the bigger amount. The option 'no change' is put there for the hurried student. The rule x^2/100 covers every version of this trap - 10% and 10% give a 1% fall, 25% and 25% give a 6.25% fall.

Ratio and proportion

A ratio is a recipe for splitting. If an amount is divided in the ratio 2 : 3 : 7, count the parts (12), find the value of one part, and multiply. When the question gives you two ratios that overlap in one letter - A : B and B : C - you cannot read off A : C directly; you must first scale both ratios so that B is the same number in each. That single step is the whole difficulty of the topic.

Ratio and proportion formulas

Term	Definition	Formula
Share of a part	Divide the total in the ratio	(own part / sum of parts) x total
Proportion a : b :: c : d	Product of extremes = product of means	ad = bc
Mean proportional between a and b	The middle term of a : x :: x : b	x = square root of ab
Third proportional to a and b	a : b :: b : x	x = b^2/a
Fourth proportional to a, b, c	a : b :: c : x	x = bc/a
Combining a : b and b : c	Scale both until b matches	Then read off a : b : c

WORKED EXAMPLE 2. (a) Rs 5,900 is divided among A, B and C so that A : B = 3 : 4 and B : C = 5 : 6. What is A's share? (b) Two numbers are in the ratio 3 : 5; if 4 is added to each the ratio becomes 5 : 7 - what is the smaller number? (c) What is the mean proportional between 9 and 64, and the third proportional to 8 and 12?

- (a) B appears as 4 in the first ratio and 5 in the second - make it 20 in both.
- (a) A : B = 3 : 4 = 15 : 20, and B : C = 5 : 6 = 20 : 24. So A : B : C = 15 : 20 : 24.
- (a) Total parts = 15 + 20 + 24 = 59, so one part = 5900/59 = Rs 100. A's share = 15 x 100 = Rs 1,500.
- (b) Let the numbers be 3x and 5x: (3x + 4)/(5x + 4) = 5/7, so 21x + 28 = 25x + 20.
- (b) 4x = 8, x = 2, so the numbers are 6 and 10 - the smaller is 6.
- (c) Mean proportional = square root of (9 x 64) = square root of 576 = 24; third proportional = 12^2/8 = 18.

Mixture and alligation - the milk-and-water family

Alligation answers one question: in what ratio do two things of different value have to be mixed to reach a given average value. The rule is a cross-subtraction - cheaper to dearer is as (dearer minus mean) is to (mean minus cheaper). Water in a milk problem is simply an ingredient costing zero. Do not build equations for these; the cross takes five seconds.

Mixture rules - four shapes, four rules

Question shape	Rule	Note
Mix two prices to reach a mean price	cheaper : dearer = (dearer - mean) : (mean - cheaper)	Water is priced at zero

Question shape	Rule	Note
Draw y litres out of x and top up with water, n times	Liquid left = x(1 - y/x)^n	Never subtract y each time
Mix EQUAL quantities of two mixtures	Add the component fractions directly	4/5 and 3/5 milk give 7 : 3
Change a : b to a : c by adding the second liquid	Keep the first quantity fixed, recompute the second	Add the difference

WORKED EXAMPLE 3 (the 2016 and 2024 family). (a) In what ratio must water be mixed with milk costing Rs 32 per litre so that the mixture is worth Rs 24 per litre? (b) A vessel contains 100 litres of pure milk; 20 litres is drawn out and replaced by water, and the operation is performed once more - how much pure milk is left? (c) A milkman buys milk at Rs 40 per litre, adds water and sells the mixture at Rs 40 per litre, gaining 25% - what is the ratio of water to milk?

- (a) Cheaper = water at Rs 0, dearer = milk at Rs 32, mean = Rs 24.
- (a) water : milk = (32 - 24) : (24 - 0) = 8 : 24 = 1 : 3.
- (b) Use milk left = x(1 - y/x)^n with x = 100, y = 20, n = 2.
- (b) 100 x (1 - 20/100)^2 = 100 x (4/5)^2 = 100 x 16/25 = 64 litres of milk (and 36 litres of water).
- (c) He sells at Rs 40 with a 25% gain, so one litre of the mixture costs him 40/1.25 = Rs 32.
- (c) Alligation between water (Rs 0) and milk (Rs 40) at a mean of Rs 32: water : milk = (40 - 32) : (32 - 0) = 8 : 32 = 1 : 4.

ASKED IN RAS

Asked in RAS Pre 2016 and again in RAS Pre 2024 - the milk-and-water mixture. The 2016 form was the alligation ratio, the 2024 form the repeated-replacement one. Learn both the cross rule and the x(1 - y/x)^n formula; between them they cover every version RPSC has set.

Partnership

Partnership is one line: profit is shared in the ratio of capital multiplied by time. If everyone stays in for the same period the time cancels and the split is simply by capital. If one partner joins late, withdraws early or changes his investment, convert each partner's money into money-months and take the ratio of those. A working partner's salary or commission is taken out of the profit FIRST; only the balance is split.

Partnership - what to multiply

Situation	Ratio to use	Example
Same period for all	Capitals	12,000 : 16,000 = 3 : 4
Different periods	Capital x months	5000 x 12 : 6000 x 8 = 5 : 4
One partner joins after m months	Capital x (12 - m)	8000 x 12 : 9000 x 8 = 4 : 3
One partner withdraws at 6 months	Capital x 6 for him, x 12 for the other	3 x 6 : 2 x 12 = 3 : 4
Working partner with commission	Deduct the commission first, split the rest	10% of 20,000 taken out first

WORKED EXAMPLE 4. (a) A invests Rs 5,000 for 12 months and B invests Rs 6,000 for 8 months; the profit is Rs 18,000 - what is A's share? (b) A and B invest Rs 30,000 and Rs 20,000; A is the working partner and takes 10% of the profit for managing, the rest being split by capital. If the annual profit is Rs 20,000, what does A get in all?

- (a) Money-months: A = 5000 x 12 = 60,000; B = 6000 x 8 = 48,000.
- (a) Ratio = 60,000 : 48,000 = 5 : 4, so 9 parts in all.
- (a) A's share = 18000 x 5/9 = Rs 10,000.

- (b) A's management fee comes out first: 10% of 20,000 = Rs 2,000.
- (b) Balance = Rs 18,000, split in the capital ratio 30,000 : 20,000 = 3 : 2.
- (b) A gets 18000 x 3/5 = Rs 10,800, so A's total = 2000 + 10800 = Rs 12,800.

TRAP

Do not split the whole profit in the capital ratio when there is a working partner - the salary or commission is deducted before the split, and the answer that ignores it is always one of the options. Equally, 'withdraws his capital at the end of 6 months' changes the TIME to 6, not the capital.

Profit, loss, marked price and discount

Fix one rule in your head and this section is safe: profit per cent and loss per cent are always on the COST price, while discount is always on the MARKED price. A shopkeeper marks his goods above cost, then discounts the marked price, and the selling price that results is compared back to the cost price to get his profit. Three different bases in one question - that is the whole difficulty.

Profit and loss formulas

Asked	Formula	Note
Profit %	(SP - CP)/CP x 100	Always on CP
SP from CP	CP(100 + p)/100	For a loss use (100 - l)
CP from SP	100 x SP/(100 + p)	Never subtract the percentage from SP
SP from marked price	MP(100 - d)/100	Discount is on MP
Mark m% above cost, discount d%	Profit % = m - d - md/100	m=25, d=10 gives 12.5%

Asked	Formula	Note
Two successive discounts	d1 + d2 - d1d2/100	20% and 10% = 28%, not 30%
Same SP, one at +x% and one at -x%	Net LOSS of $x^2/100$ per cent	Never a break-even
CP of x articles = SP of y articles	Profit % = (x - y)/y x 100	Loss if y is bigger
False weight, sold at cost price	Gain % = error/(true value - error) x 100	800 g for 1 kg gives 25%
Profit of p% on CP, expressed on SP	100p/(100 + p) per cent of the SP	25% on cost = 20% of the selling price

The shopkeeper chain - work along it, never jump

Cost price (CP) → Mark up m% -> Marked price (MP) → Discount d% on MP -> Selling price (SP) → Compare SP with CP -> profit or loss %

WORKED EXAMPLE 5. (a) A shopkeeper buys an article for Rs 300, marks it 25% above cost and allows a discount of 10% on the marked price - what is his profit per cent? (b) A man sells two articles at Rs 1,200 each, gaining 20% on one and losing 20% on the other - what is his overall result? (c) A dishonest dealer sells at cost price but uses an 800 g weight for a kilogram - what is his gain per cent?

- (a) Marked price = 300 x 1.25 = Rs 375.
- (a) Selling price = 375 x 0.90 = Rs 337.50, so profit = Rs 37.50 on a cost of Rs 300.
- (a) Profit % = 37.5/300 x 100 = 12.5%. Shortcut: m - d - md/100 = 25 - 10 - 2.5 = 12.5%.
- (b) CP of the first = 1200 x 100/120 = Rs 1,000; CP of the second = 1200 x 100/80 = Rs 1,500.

- (b) Total CP = Rs 2,500 against a total SP of Rs 2,400 - a LOSS of Rs 100, i.e. 4% = x^2/100 with x = 20.
- (c) Gain % = error/(true value - error) x 100 = 200/800 x 100 = 25%.

TRAP

Successive discounts do NOT add. Discounts of 20% and then 10% on Rs 1,000 give 1000 x 0.80 x 0.90 = Rs 720, an effective discount of 28% - not 30%. The single equivalent discount is d1 + d2 - d1d2/100, and it is always LESS than the sum. The same warning applies to two successive profit percentages.

Simple interest

Simple interest is charged on the original principal every year, so the interest for each year is identical. That single property answers most questions: if a sum amounts to different values after different numbers of years, the difference between the two amounts is the interest for the gap in years, and dividing gives the interest for one year.

Simple interest - one formula, four rearrangements

Asked	Formula	Trick version
Interest	SI = P x R x T/100	Amount = P + SI
Principal	P = SI x 100/(R x T)	-
Rate	R = SI x 100/(P x T)	-
Time	T = SI x 100/(P x R)	-
Sum doubles itself	SI = P, so R x T = 100	R = 100/T
Sum triples itself	SI = 2P, so R x T = 200	R = 200/T
Amounts known for two times	Difference = interest for the gap	Then work back to P

WORKED EXAMPLE 6. (a) At what rate of simple interest will a sum double itself in 8 years? (b) A sum at simple interest amounts to Rs 2,400 in 3 years and to Rs 2,800 in 5 years - what is the sum? (c) At what rate will a sum become three times itself in 20 years?

- (a) To double, the interest must equal the principal, so SI = P.
- (a) P = P x R x 8/100, giving R = 100/8 = 12.5% per annum.
- (b) Interest for the 2-year gap = 2800 - 2400 = Rs 400, so interest for one year = Rs 200.
- (b) Interest for 3 years = Rs 600, so the principal = 2400 - 600 = Rs 1,800.
- (c) To triple, the interest must be twice the principal: 2P = P x R x 20/100.
- (c) R = 200/20 = 10% per annum.

Compound interest, growth, depreciation and instalments

Compound interest adds the interest back to the principal, so the base grows every year. Everything else follows from the one formula A = P(1 + R/100) raised to the power n. Population growth is the same formula; depreciation is the same formula with a minus sign. For half-yearly compounding halve the rate and double the number of periods; for quarterly, quarter the rate and take four times the periods.

Compound interest and its cousins

Situation	Formula	Example
Amount after n years	A = P(1 + R/100)^n	CI = A - P
Half-yearly compounding	Rate R/2, periods 2n	10% for 1 year becomes 5% for 2 periods
Quarterly compounding	Rate R/4, periods 4n	-

Situation	Formula	Example
Population after n years	P(1 + r/100)^n	20,000 at 10% for 2 years = 24,200
Depreciation	P(1 - r/100)^n	50,000 at 10% for 2 years = 40,500
Doubling at CI	Doubles in t, so 4 times in 2t, 8 times in 3t	Doubles in 5 years -> 4 times in 10
Rate from a known amount	(1 + R/100)^n = A/P	1.44 over 2 years gives R = 20%
Different rates in different years	Multiply the separate factors	P(1 + r1/100)(1 + r2/100)

WORKED EXAMPLE 7. (a) What is the compound interest on Rs 8,000 for 1 year at 10% per annum, compounded half-yearly? (b) If Rs 6,000 fetches a compound interest of Rs 2,640 in 2 years, what is the rate? (c) A sum doubles itself in 5 years at compound interest - in how many years does it become four times?

- (a) Half-yearly: use 5% for 2 periods, not 10% for 1 year.
- (a) Amount = 8000 x (1.05)^2 = 8000 x 1.1025 = Rs 8,820, so CI = Rs 820 (annual compounding would give only Rs 800).
- (b) Amount = 6000 + 2640 = Rs 8,640, so (1 + R/100)^2 = 8640/6000 = 1.44.
- (b) 1 + R/100 = 1.2, hence R = 20% per annum.
- (c) In 5 years P becomes 2P; in the next 5 years that 2P doubles again to 4P.
- (c) So 5 + 5 = 10 years. Becoming 2 to the power k times takes 5k years.

Instalments at compound interest – the present-value method

Call each instalment x → Discount the first year's instalment: $x/(1 + r/100)$ → Discount the second year's: $x/(1 + r/100)^2$ → Add all the discounted values → Set the sum equal to the debt and solve for x

WORKED EXAMPLE 8. A debt of Rs 2,100 is to be repaid in two equal annual instalments at 10% per annum compound interest. What is each instalment?

- Each instalment is x; the first is paid after 1 year, the second after 2 years.
- Present value = $x/1.1 + x/(1.1)^2 = 2100$.
- Take the common denominator 1.21: $x(1.1 + 1)/1.21 = 2100$, i.e. $x \times 2.1/1.21 = 2100$.
- $x = 2100 \times 1.21/2.1 =$ Rs 1,210.
- Check: $1210/1.1 =$ Rs 1,100 and $1210/1.21 =$ Rs 1,000, and $1100 + 1000 =$ Rs 2,100. Correct.
- Note the simpler instalment question: cash price Rs 12,000 against 4,000 down plus two yearly payments of 4,200 means 12,400 paid in all, i.e. Rs 400 extra.

The difference between compound and simple interest – four papers running

This is the single most repeated question in the whole reasoning paper – RAS Pre 2018, 2021, 2023 and 2024, four consecutive papers. Understand why the difference exists and you will never need to compute both interests. Simple interest charges the same interest every year. Compound interest, from the second year onwards, also charges interest on the previous year's interest. That extra bit IS the difference. For two years the extra bit is one year's interest on the first year's interest, which is exactly P(R/100) squared.

Simple interest vs compound interest

Simple interest	Compound interest
Interest on the original principal only	Interest on principal plus accumulated interest
Same interest every year	Interest grows every year
SI = P x R x T/100	A = P(1 + R/100)^n and CI = A - P
Equal for the first year	Equal for the first year - the gap opens in year 2
Doubles when R x T = 100	Doubles in t years, quadruples in 2t
Grows in a straight line	Grows faster and faster; CI always exceeds SI for n > 1

The difference formulas - memorise all four lines

Asked	Formula	Worked value
Difference for 2 years	D = P(R/100)^2 = P x R^2/10000	P=10,000, R=5: Rs 25
Difference for 3 years	D = P x R^2 x (300 + R)/10^6	P=16,000, R=5: Rs 122
Same, other form	D = P(R/100)^2 x (3 + R/100)	-
Find the sum from a 2-year difference	P = D x (100/R)^2	D=32, R=4: Rs 20,000
Find the rate from a 2-year difference	R = 100 x square root of (D/P)	D=144, P=10,000: 12%
Find the sum from a 3-year difference	P = D x 10^6 / (R^2 x (300 + R))	D=62, R=10: Rs 2,000

WORKED EXAMPLE 9 (the two-year type). (a) What is the difference between the compound interest and the simple interest on Rs 10,000 for 2 years at 5%

per annum? (b) The same for Rs 8,000 for 2 years at 10% per annum. (c) The same for Rs 25,000 for 2 years at 8% per annum.

- (a) Long way: SI = 10000 x 5 x 2/100 = Rs 1,000; CI = 10000 x (1.05)^2 - 10000 = 11,025 - 10,000 = Rs 1,025.
- (a) Difference = 1025 - 1000 = Rs 25. Shortcut: P(R/100)^2 = 10000 x (0.05)^2 = 10000 x 0.0025 = Rs 25.
- (b) P(R/100)^2 = 8000 x (0.1)^2 = 8000 x 0.01 = Rs 80. (Check: SI 1,600, CI 1,680.)
- (c) P(R/100)^2 = 25000 x (0.08)^2 = 25000 x 0.0064 = Rs 160. (Check: SI 4,000, CI 4,160.)
- Notice the pattern - the shortcut needs one multiplication, the long way needs four.

WORKED EXAMPLE 10 (three years, and the reverse questions). (a) Find the difference on Rs 16,000 for 3 years at 5% per annum. (b) The difference for 2 years at 4% per annum is Rs 32 - find the sum. (c) The difference on Rs 10,000 for 2 years is Rs 144 - find the rate. (d) The difference for 3 years at 10% is Rs 62 - find the sum.

- (a) D = P x R^2 x (300 + R)/10^6 = 16000 x 25 x 305/1,000,000 = Rs 122.
- (a) Check: SI = Rs 2,400 and CI = 16000 x (1.05)^3 - 16000 = 18,522 - 16,000 = Rs 2,522. Difference Rs 122.
- (b) 32 = P x (4/100)^2 = P x 0.0016, so P = 32/0.0016 = Rs 20,000.
- (c) 144 = 10000 x (R/100)^2, so (R/100)^2 = 0.0144 and R/100 = 0.12, giving R = 12%.
- (d) D = P x 100 x 310/1,000,000 = 0.031P, so 0.031P = 62 and P = Rs 2,000.

- For 10% over 3 years on Rs 10,000 the difference is Rs 310 - worth remembering as a benchmark.

ASKED IN RAS

Asked in RAS Pre 2018, 2021, 2023 and 2024 - four consecutive papers. Both the direct form (find the difference) and the reverse form (the difference is given, find the sum or the rate) have appeared. If you learn one formula from this whole chapter, learn $D = P(R/100)^2$ for two years and $D = P \times R^2 \times (300 + R)/10^6$ for three.

REVISION RECAP

1. Difference between CI and SI for 2 years = $P(R/100)^2$; for 3 years = $P \times R^2 \times (300 + R)/10^6$.
2. Reverse the same formulas: $P = D \times (100/R)^2$, and $R = 100 \times$ the square root of D/P.
3. Benchmarks worth carrying: 10,000 at 5% for 2 years gives Rs 25; 10,000 at 10% for 3 years gives Rs 310.
4. $SI = PRT/100$ and it is the same every year; a sum doubles when $R \times T = 100$ and triples when $R \times T = 200$.
5. CI: $A = P(1 + R/100)^n$. Half-yearly means rate $R/2$ and periods $2n$; quarterly means $R/4$ and $4n$.
6. At compound interest a sum that doubles in t years becomes 4 times in $2t$ and 8 times in $3t$.
7. Population growth is $P(1 + r/100)^n$ and depreciation the same with a minus sign.
8. Instalments at CI: discount each instalment back to today and set the total equal to the debt.
9. Divide in a ratio by parts; to join A : B with B : C, scale both until B matches, then read A : B : C.
10. Mean proportional = square root of ab, third proportional = b^2/a, fourth proportional = bc/a.
11. Alligation: cheaper : dearer = (dearer - mean) : (mean - cheaper), with water priced at zero.
12. Repeated replacement leaves $x(1 - y/x)^n$ of the original liquid - never subtract y each time.
13. Partnership shares profit in the ratio of capital x time; the working partner's commission comes out first.

14. Successive percentage changes combine as a + b + ab/100, never by simple addition.
15. An x% rise followed by an x% fall is always a net fall of x^2/100 per cent.
16. A price rise of x% needs a consumption cut of 100x/(100 + x) per cent to hold expenditure steady.
17. Profit and loss percentages sit on the cost price; discount sits on the marked price.
18. Mark up m% and discount d% gives a profit of m - d - md/100 per cent.
19. Two discounts d1 and d2 equal a single discount of d1 + d2 - d1d2/100 - always less than the sum.
20. Selling two articles at the same price with equal gain and loss percentages always loses x^2/100 per cent.